

Ex.No. 14: 5G Throughput vs SNR with Modulation Labels Date

Objective 1 Matlab Code and Output:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation
Throughput = Max_TP*(1-exp(-SNR/10));
disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,Throughput,'-o','LineWidth',2)
title('5G Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2 -----

subplot(2,1,2)
bar(Throughput)
title('Throughput at Different SNR Levels')
```

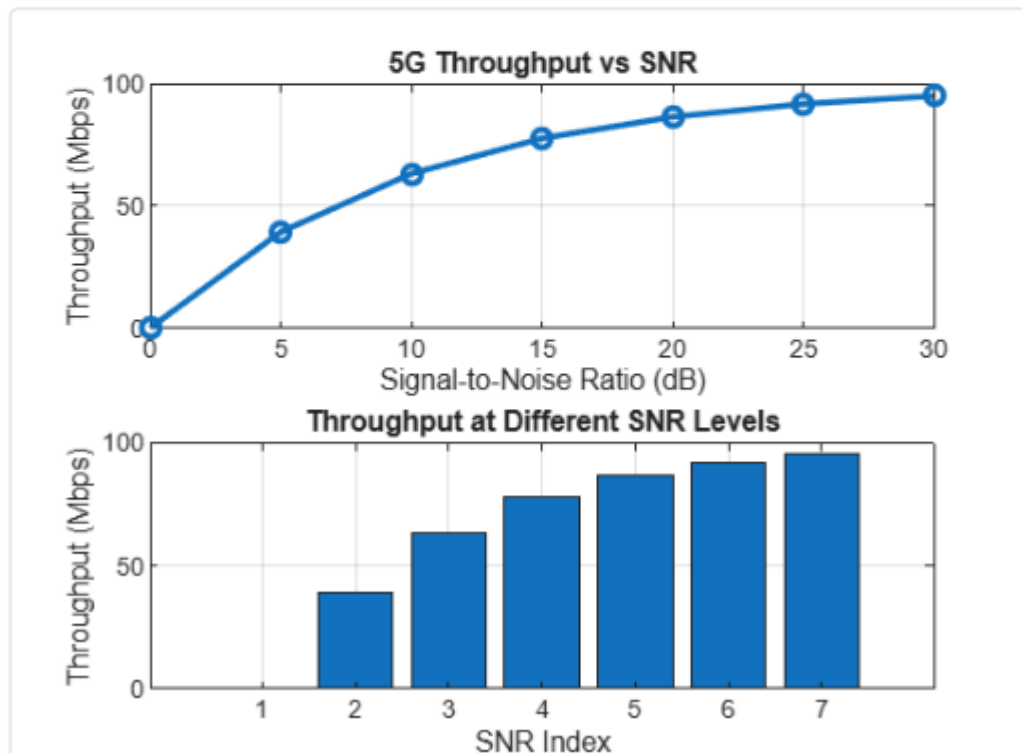
```
xlabel('SNR Index')
```

```
ylabel('Throughput (Mbps)')
```

```
grid on
```

```
SNR(dB) Throughput(Mbps)
```

	0	0
5.0000	39.3469	
10.0000	63.2121	
15.0000	77.6870	
20.0000	86.4665	
25.0000	91.7915	
30.0000	95.0213	



Objective 2 Matlab Code and Output:

```
clc;
```

```
clear;
```

```
close all;
```

```
% SNR values (dB)
```

```
SNR = 0:5:30;
```

% Simulated Bit Error Rate (BER)

BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];

% Reliability (%)

Reliability = (1 - BER) * 100;

% Display values

disp('SNR(dB) BER Reliability(%)');

disp([SNR' BER' Reliability']);

figure;

% ----- Graph 1: SNR vs BER -----

subplot(2,1,1);

semilogy(SNR, BER, '-o', 'LineWidth', 2);

title('SNR vs Bit Error Rate in 5G');

xlabel('Signal-to-Noise Ratio (dB)');

ylabel('Bit Error Rate (BER)');

grid on;

% ----- Graph 2: SNR vs Reliability -----

subplot(2,1,2);

plot(SNR, Reliability, '-s', 'LineWidth', 2);

title('SNR vs Communication Reliability');

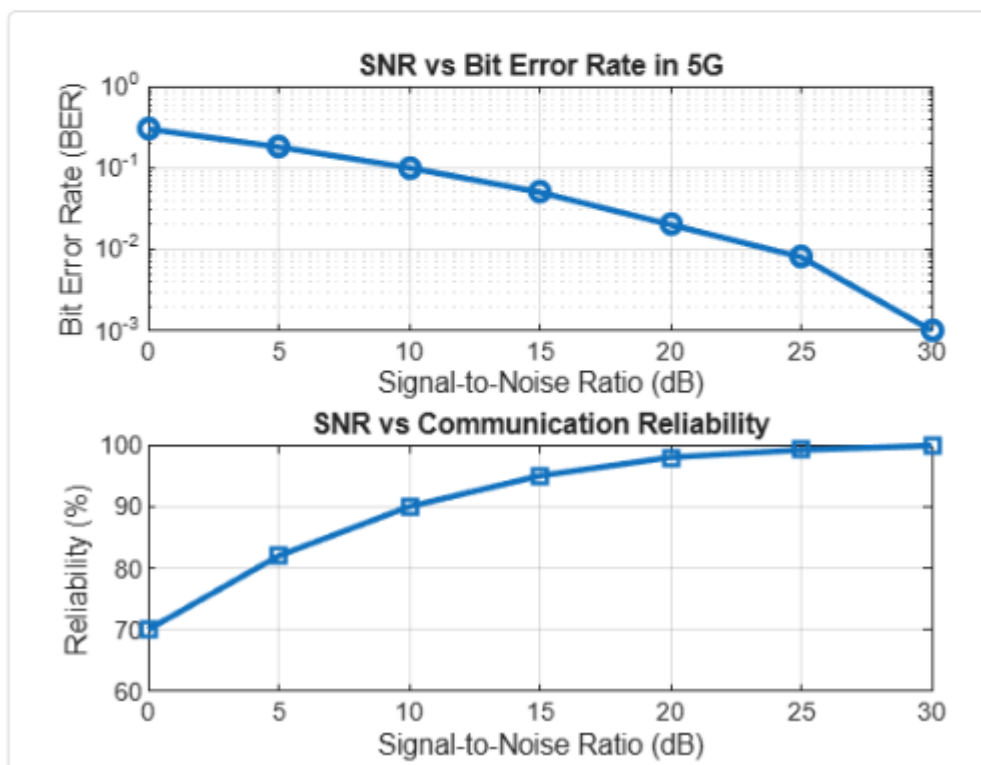
xlabel('Signal-to-Noise Ratio (dB)');

```
ylabel('Reliability (%)');
```

```
grid on;
```

```
ylim([60 100]);
```

SNR(dB)	BER	Reliability(%)
0	0.3000	70.0000
5.0000	0.1800	82.0000
10.0000	0.1000	90.0000
15.0000	0.0500	95.0000
20.0000	0.0200	98.0000
25.0000	0.0080	99.2000
30.0000	0.0010	99.9000



Objective 3 Matlab Code and Output: `clc;`

`clear;`

`close all;`

% Modulation schemes

Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};

% Required SNR (dB)

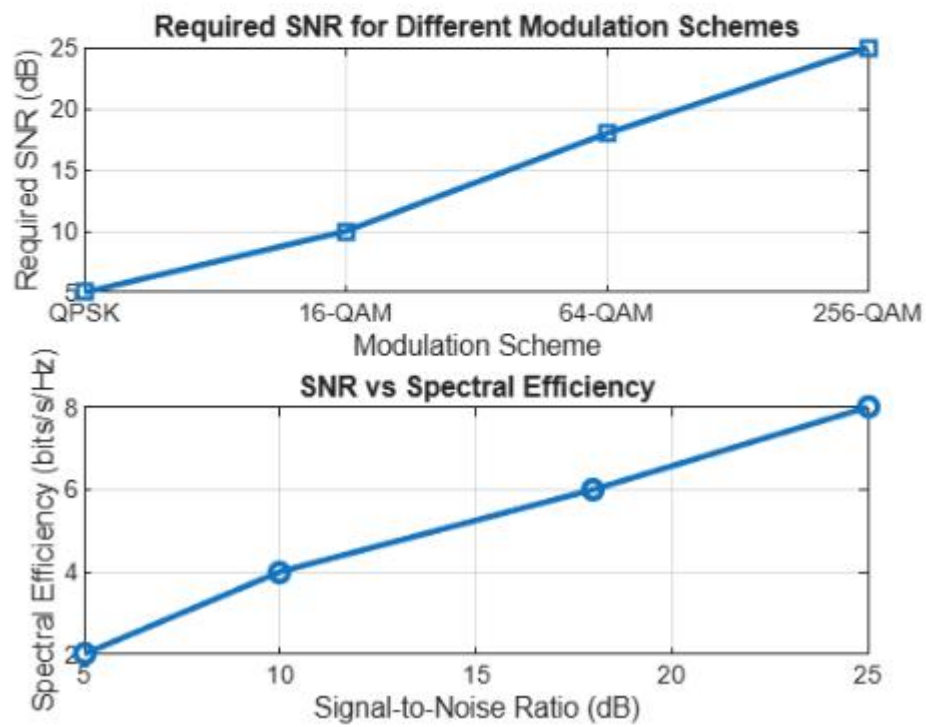
```

SNR = [5 10 18 25];
% Spectral Efficiency (bits/s/Hz)
Efficiency = [2 4 6 8];
disp('Required SNR (dB) and Spectral Efficiency')
disp(table(Modulation',SNR',Efficiency', ...
'VariableNames',{'Modulation','SNR_dB','Efficiency'}))
figure
% ----- Graph 1 -----
subplot(2,1,2)
plot(SNR,Efficiency,'-o','LineWidth',2)
title('SNR vs Spectral Efficiency')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Spectral Efficiency (bits/s/Hz)')
grid on
% ----- Graph 2 -----
subplot(2,1,1)
plot(1:4,SNR,'-s','LineWidth',2)
xticks(1:4)
xticklabels(Modulation)
title('Required SNR for Different Modulation Schemes')
xlabel('Modulation Scheme')
ylabel('Required SNR (dB)')

```

Required SNR (dB) and Spectral Efficiency

Modulation	SNR_dB	Efficiency
{ 'QPSK' }	5	2
{ '16-QAM' }	10	4
{ '64-QAM' }	18	6
{ '256-QAM' }	25	8



grid on